Harsh Gupta

930-904-3651 | harsh.gupta34@outlook.com | linkedin.com/in/harshg1347 | harsh1347.github.io

Summary

Data Engineer with an MS in Data Science and 3+ years of production experience designing ETL/ELT pipelines, data ingestion frameworks, and enterprise data warehouse solutions across AWS and Azure. Skilled in PySpark, SQL, AWS Glue, Airflow, Azure Data Factory, Redshift, OracleDB, and Postgres, with experience building FSI-domain data systems at Deloitte and ML-integrated pipelines at Indiana University. Strong foundation in dimensional modeling, workflow orchestration, data validation, reconciliation, lineage, and cloud-native pipeline engineering.

Education

Indiana University Bloomington Aug 2024 – May 2026
Master of Science in Data Science (GPA: 4.0/4.0) *Bloomington, IN*

Manipal University Jul 2017 – May 2021
B.Tech in Electronics and Communication, Minor in Data Science (GPA: 8.13/10) *Manipal, India*

Technical Skills

Programming: Python, PySpark, SQL, R
Cloud & Data Platforms: AWS, Azure, Redshift, OracleDB, Postgres, Amazon Aurora, Hadoop, Tableau, PowerBI
Data Engineering: AWS Glue, Spark, Airflow, dbt, Kafka, Snowflake, ETL/ELT Pipelines, Data Ingestion, Data Modeling, Data Warehouse Design, Data Quality, Data Validation, Reconciliation, SLA Monitoring, Docker, CI/CD
Generative AI: LangChain, LangGraph, RAG, Azure AI Studio, Hugging Face

Full-Time Experience

Deloitte – Data & AI Consultant Sep 2021 – Jul 2024
Data Engineer – FSI Domain | AWS Glue, PySpark, Spark, OracleDB, Postgres *Bengaluru, India*

- Engineered nightly batch ETL pipelines in AWS Glue and PySpark to integrate a new XML-based insurance source into the existing **Landing** $\rightarrow$ **Staging** $\rightarrow$ **ODS** architecture, processing **12–15M records per batch** and delivering data within **SLA** for downstream analytics.
- Designed **fact and dimensional table models** in the ODS layer to serve **policy analysis, fund analysis, and reporting** workloads; integrated new source outputs with existing Informatica workflows and downstream analytics tables.
- Orchestrated pipelines using AWS Glue Workflows with EventBridge scheduling and SNS alerting, reducing issue resolution time by **25–30%**; implemented row-level audit tracking, reconciliation logic, and reusable transformation frameworks ensuring **data lineage and quality** at scale.

Data Scientist – Customer Strategy & Pricing | Python, R, ML, AWS S3, Tableau

- Modeled demand elasticity across 5,000+ store locations, increasing profit by **$1M–$3M**; streamlined the Pricing-as-a-Service data pipeline, improving efficiency by **30%**.

Firm Initiative | PySpark, Airflow, Flask, LangChain, Postgres, AWS S3

- Built **CodETL**, a platform-agnostic ETL engine with Airflow-orchestrated topologically sorted schedules enabling 25+ transformations and **reducing development effort by 40%**, and **DataLens**, an LLM-powered RAG-based Q&A portal for natural language querying over structured and unstructured enterprise data.
- Presented CodETL at the **Deloitte AI & DE Summit**; earned **4+ firm awards** for successful delivery of 3 internal systems.

Part-Time & Internship

Indiana University Bloomington Oct 2024 – Present
Part-Time AI & Data Engineer | Azure Data Factory, Python, SQL, Tableau *Bloomington, IN*

- Led development of an end-to-end time-series forecasting pipeline in Python and Azure Data Factory, extracting dining transaction data, engineering ML features, and delivering daily resource allocation predictions to dining managers – improving **prediction accuracy by 21%**.
- Accelerated **pipeline throughput 6$\times$** using distributed Azure compute, reducing **residual forecast error to 5%** across millions of dining transaction records.

Indiana University Bloomington Campus Auxiliaries Jun 2025 – Aug 2025
AI & Data Engineer Intern | Azure AI Search, Azure ML, Python, FastAPI, LangChain *Bloomington, IN*

- Designed and built a document ingestion and indexing pipeline with Python and Azure services that onboarded **12,000+ Confluence pages** into Azure AI Search, automating extraction, chunking, embedding, and index updates, allowing users to search newly added content **within minutes of ingestion**.
- Improved retrieval accuracy from **76% to 93%** via metadata filtering and reranking; deployed a RAG-based retrieval system enabling 250+ staff to reduce document lookup time from **20–25 min to under 10 min**.